LUCIDMSK CME PROGRAM

MODULE 06

Wrist & Hand MRI: Ligaments, Tendons & Classification Systems

SL Ligament · TFCC Palmer · Kienbock · Stener · ECU · Flexor Pulley · CMC OA

1.0 AMA PRA Cat 1 | **Wrist/Hand** | **CME Article Review**

LEARNING OBJECTIVES

1. Apply Palmer classification to TFCC tears (Types 1A–1E, 2A–2E).
2. Grade scapholunate ligament injuries using the Geissler classification.
3. Recognize and classify thumb UCL injuries including the Stener lesion.
4. Apply the Lichtman classification to Kienbock disease staging.
5. Classify first CMC osteoarthritis using the Eaton-Littler system.
6. Grade flexor pulley injuries and quantify bowstringing on axial MRI.

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1. TFCC Injuries — Palmer Classification

Palmer Classification — TFCC Tears

Type	Subtype	Description	Repairability
1 (Traumatic)	1A — Central disc	Horizontal perforation central articular disc	Non-repairable; debridement
1 (Traumatic)	1B — Peripheral ulnar	Detachment at ulnar styloid / fovea	Best repair candidate; arthroscopic suture
1 (Traumatic)	1C — Distal	Tear of ulnocarpal ligaments	Repair if DRUJ unstable
1 (Traumatic)	1D — Radial	Avulsion from sigmoid notch	Repair; associated DRUJ instability
2 (Degenerative)	2A–2E	Progressive chondromalacia → perforation → LT tear → arthrosis	USO if + ulnar variance

KEY POINT

1B (peripheral ulnar) tears detach at the fovea and are the most surgically important — best repair candidates. Type 2 degenerative tears are managed with ulnar shortening osteotomy when + ulnar variance.

2. Scapholunate, Stener Lesion & Kienbock

Geissler SL Grading (Arthroscopy / MRI Correlation)

Grade	MRI Finding	Arthroscopy	Instability
I	Intact; minor signal change	Attenuation; no probe	None
II	Partial tear; no malalignment	Probe palpation; no passage	Minimal
III	Partial/complete; SL gap	Probe passes through gap	Step-off; laxity
IV	Complete; DISI alignment	Scope (2.7 mm) passes freely	Gross instability

Stener Lesion — UCL Injury Classification (1st MCP)

Grade	Pathology	MRI Finding	Management
I — Sprain	Intact fibers	Edema; continuous	Thumb spica 4–6 weeks
II — Partial	Partial disruption	Partial gap; deep to aponeurosis	Splinting; consider repair >50%
III — Complete (non-Stener)	Full rupture; stump deep	Complete gap; deep to aponeurosis	Conservative possible

Grade	Pathology	MRI Finding	Management
STENER	Full rupture; stump SUPERFICIAL	Stump superficial; "yo-yo" sign	Surgical repair REQUIRED

Lichtman Classification — Kienbock Disease

Stage	Radiograph	MRI	Management
I	Normal	T1 hypointense; no fracture	Immobilization; conservative
II	Sclerosis; cysts; no collapse	Established AVN	Joint leveling; radial shortening
IIIA	Collapse; NO fixed scaphoid rotation	Reducible scaphoid rotation	Revascularization; limited carpal procedures
IIIB	Collapse + FIXED DISI (scaphoid rotation)	Fixed DISI deformity	Scaphocapitate fusion; salvage
IV	Radiocarpal arthrosis	Pancarpal arthrosis	PRC or total wrist fusion

> **CRITICAL**
>
> IIIA vs. IIIB is the critical surgical decision: IIIB has FIXED scaphoid rotation (DISI) changing the procedure from revascularization to carpal stabilization surgery. Always assess scaphoid rotation on CT/MRI.

References

- Palmer AK. J Hand Surg 1989.
- Stener B. J Bone Joint Surg Br 1962.
- Lichtman DM et al. J Hand Surg 1977.

3. Eaton-Littler CMC OA & Flexor Pulley Injuries

Eaton-Littler CMC Osteoarthritis

Stage	Radiographic Findings
I	Normal joint space (≤1 mm narrowing); cartilage preserved
II	Narrowing + sclerosis + osteophytes; subluxation <1/3 joint width
III	Subluxation >1/3 + larger osteophytes (>2 mm); no pantrapezial OA
IV	Pantrapezial arthrosis (STT + CMC involved); severe changes

Flexor Pulley Injury Grading (Axial MRI)

Grade	Finding	Bowstringing
1 — Sprain	Pulley thickening; fibers intact	None (<1 mm)

Grade	Finding	Bowstringing
2 — Partial	Partial fiber disruption; attenuated	<3 mm
3 — Complete (single)	Full-thickness single pulley rupture	> 3 mm (pathognomonic)
4 — Multiple	A2 + A3 or A2 + A4 complete tears	Severe; surgical reconstruction

References

- Eaton RG, Littler JW. J Bone Joint Surg Am 1973.
- Geissler WB et al. J Hand Surg 1996.